University of Macau

Faculty of Information Science and Computing



澳門大學

UNIVERSIDADE DE MACAU

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CISC7204: Data Science and Data Visualization

Class E (2026-2027 Semester 1)

Team Project Proposal

Visualizing the Relationship Between Visitor Arrivals, Hotel Occupancy, and Bus Ridership in Macao

Group name: Class E - Team 01

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1. Project Title and Research Question

Title: Visualizing the Relationship Between Visitor Arrivals, Hotel Occupancy, and Bus Ridership in Macao

Macao is a small and dense city where tourism and daily transport share the same space. When many visitors arrive, hotels may fill up and buses may become busier. This project checks this idea with monthly data and clear visualizations. The data work is already far enough advanced to report a first result for every research question: Section 4 shows the cleaned files, the figures and the coefficients behind them.

Research questions

- RQ1 (synchrony): Did monthly visitor arrivals, hotel occupancy rate, and bus ridership in Macao move together during 2023-2025?
- RQ2 (seasonality): How do the seasonal patterns of the three indicators differ across the four quarters (Q1 = months 1-3, Q2 = 4-6, Q3 = 7-9, Q4 = 10-12)? These quarters contain Lunar New Year, Labour Day, the summer holidays, and National Day Golden Week.

Hypotheses

- H1: Visitor arrivals and bus ridership are positively correlated.
- H2: Visitor arrivals and hotel occupancy are positively correlated. The relationship is expected to be stronger for overnight visitors than for same-day visitors.
- H3: Hotel occupancy and bus ridership remain associated after considering visitor arrivals (partial correlation or a controlled view). We acknowledge that both may be driven by arrivals, so H3 tests only whether any association is left.

Limitations to state in the final report

- Correlation is not causation.
- Bus ridership includes local residents, not only visitors.
- The sample is small: 36 months, or 12 quarters. Quarterly aggregation makes those twelve observations readable, but it does not create information, so every coefficient is reported with its window and its n.
- 2023 contains a recovery trend that may inflate correlations. The report therefore also gives the coefficient for 2024-2025 alone and for each year separately, and discusses the difference instead of pooling it away.
- The supply side is one timetable snapshot rather than a monthly series: the stop-level file carries scheduled weekday, weekend and holiday departures from the current DSAT timetable, so Figures 2 and 3 test where and when capacity is scheduled, not how it moved month by month, and any change in supply inside the window is invisible to them.

Analytical decisions that follow from these limitations

- Two windows, two different jobs. The general trend is read on the long run, January 2015 to August 2026, so that the pandemic collapse and the 2023-2025 rebuilding appear as history rather than as an unexplained gap. Every statistic behind RQ1, RQ2 and H1-H3 is computed on the 36 months from January 2023 to December 2025. Figure 1 draws both on one axis: the long line supplies context, the shaded band marks the months that are actually estimated.
- The pandemic years are excluded from the arithmetic, not only from the narrative. No mean, index or coefficient is computed on 2020-2022: monthly arrivals averaged 3.28 million in 2019, fell to 11,041 in April 2020, and only returned to the 2019 level in 2025, so a statistic computed across that collapse would measure border policy.
- The quarter is the reporting unit. Thirty-six months is twelve quarters, which is the resolution the seasonality question needs, so the quarterly table and the quarterly coefficient are reported next to the monthly ones; a reader who thinks in quarters never has to convert anything.

- A short window is defended by splitting it, not by hiding it. Coefficients are reported for the pooled 36 months, for 2024-2025 where the level is stable, and for each year separately. A relationship that exists only in 2023 is labelled a recovery effect, not a finding.
- Colour and shape carry the meaning together. Every figure uses a colourblind-safe palette and also separates series by marker, line style and direct labels, so the same figure stays readable in grayscale print, in a photocopy, and for a reader with colour-vision deficiency.

2. Project Purpose

Learning outcomes

- Build a full data science workflow: acquire, clean, merge, explore, analyse, visualize, and report.
- Choose chart types that fit the question (comparison, relation, composition, distribution).
- Judge the quality of a chart using Tufte's data-ink ratio and lie factor (Tufte, 2001) and the ideas of truthful visualization (Cairo, 2016).
- Communicate findings clearly to a non-technical reader (Knaflic, 2015).

Skills

- Python and pandas: reading files, cleaning, to_datetime, merge, groupby, pivot and melt (McKinney, 2022).
- Matplotlib and Seaborn: twin-axis time series, shaded event periods, correlation heatmap, boxplots (Hunter, 2007; Hunter & the Matplotlib development team, 2024; Waskom, 2021).
- Basic statistics: Pearson correlation, partial correlation, kernel density estimation (Downey, 2014; Virtanen et al., 2020).
- A basic interactive dashboard in Plotly Dash, built around one shared time axis, a district filter and an hour slider.
- Teamwork: shared repository, meeting minutes, task tracking, and peer review.

Disciplinary knowledge

- How tourism demand connects to hotel use and urban public transport in a tourism-based city.
- How holidays and events shape seasonal patterns.
- How to read official statistics and their definitions with care.

3. Data Context

Item	Description
Region	Macao SAR, China
Domain	Tourism and urban transport
Main analysis period	January 2023 to December 2025 (36 months, 12 quarters). This is the only window used for statistics; 2020-2022 is excluded because arrivals and occupancy in those years were set by border policy rather than by demand.
Supplementary period	2026 up to the latest published month. Used only to extend the district-hour and departure layers, never to compute a correlation or a seasonal index.
Context chart	Figure 1 plots monthly arrivals from 2015 to 2026 with the pandemic years left blank, and shades the 36-month core window inside it. Its job is context and event timing, not estimation.

Analysis window, the pandemic years, and the quarterly view

The window is fixed at January 2023 to December 2025: 36 months, 12 quarters, 103.2 million visitor arrivals and 42.6 million hotel guests. The three monthly series only overlap from January 2020 onwards, so the

binding constraint is not availability but distortion: 2023-2025 is the first span in which the level of arrivals, occupancy and departures is set by demand rather than by border policy, which is why estimation stops at December 2025 while the 2026 months are carried for context.

Figure 1 keeps context and estimation separate. The long line runs from 2015 to 2026 with 2020-2022 left blank: the fall from 3.28 million arrivals a month in 2019 to 11,041 in April 2020, and the return to 3.34 million in 2025, is why those three years cannot be pooled with the rest. The shaded band marks the 36 months every number here comes from.

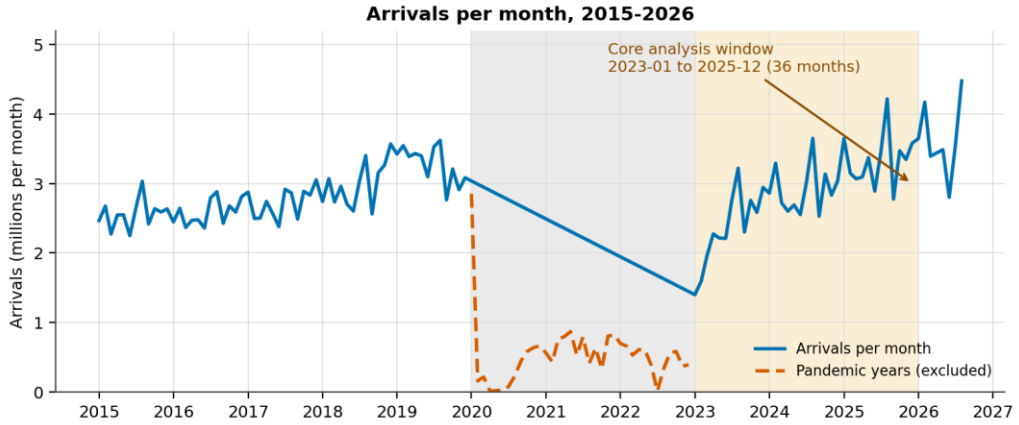


Figure 1. Just under twelve years of inbound volume, with the crisis years kept visible but excluded from every calculation. The red dashed segment marks those three years; the shaded band is the estimation period; the vertical axis counts millions of arrivals. Source: Statistics and Census Service (n.d.); team calculation.

Event analysis follows the same two-step logic. Each event - Lunar New Year, Labour Day, summer, Golden Week - is first placed on the long-run line of Figure 1, so that its size can be judged against a normal year rather than against the recovery period; only then is its window measured inside the 36 months, by comparing the event weeks with the same weeks of the other years in the window. A festival that looks large in 2023 may simply be the rebuilding of the market, and the general trend is what separates the two readings.

Inside the window the three indicators move in the same direction but at different speeds. Quarterly arrivals rise from 4.95 million in 2023Q1 to 10.40 million in 2025Q4, hotel occupancy from 74.9 to 89.9 per cent, and average daily bus trips from 549,000 to 659,000. Supply therefore grows by about 20 per cent while demand more than doubles, and bus trips per visitor fall from 9.96 to 5.83, a decline of 41 per cent. That ratio is the central quantity of the project: it is what the dashboard has to make visible, and exactly what a headline correlation of arrivals against ridership would hide. Table 1 gives the twelve quarterly observations the seasonality work will use.

Table 1. Quarterly aggregation of the 36-month window. Arrivals and hotel guests are sums within the quarter, occupancy is the monthly average, bus trips are the average of daily trips, and the last column is quarterly passenger trips divided by quarterly arrivals. Source: Statistics and Census Service (n.d.) for arrivals, guests and occupancy; DSAT transport statistics (Transport Bureau, n.d.) for bus trips; team calculation.

Quarter	Visitor arrivals (million)	Hotel occupancy (%)	Bus trips per day (thousand)	Bus trips per visitor
2023 Q1	4.95	74.9	549	9.96
2023 Q2	6.70	80.7	583	7.92
2023 Q3	8.28	85.4	601	6.67
2023 Q4	8.28	83.9	616	6.83
2024 Q1	8.88	84.9	613	6.29
2024 Q2	7.84	83.1	609	7.07
2024 Q3	9.20	88.4	635	6.35
2024 Q4	9.01	89.0	641	6.55
2025 Q1	9.86	90.2	629	5.74
2025 Q2	9.36	88.1	625	6.08
2025 Q3	10.45	89.5	632	5.56
2025 Q4	10.40	89.9	659	5.83

Datasets

ID	Dataset	Source / link	Status
3.1	Visitor arrivals by source market (focus: Mainland China, Hong Kong, Taiwan; the rest grouped as Others)	Macao Tourism Data Plus: https://dataplus.macaotourism.gov.mo/Indicator/VisitorsInEntryPointNPlace/TotalBar?lang=E	Raw, not cleaned; the M1 lead cleans it in week 1
3.2	Visitor arrivals by overnight / same-day	Macao Tourism Data Plus (same portal, https://dataplus.macaotourism.gov.mo/). Exact indicator page to be recorded by M1 lead.	Cleaned; monthly same-day and overnight series, 2008-01 to 2026-08
3.3	Occupancy rate of operating hotel establishments (monthly)	Macao Tourism Data Plus (main source). DSEC (https://www.dsec.gov.mo/en-US/) may be used as a supplement to cross-check.	Cleaned; 45 months (Jan 2023 - Sep 2026); the 36-month window is used for estimation
3.4	Bus ridership (monthly: average daily trips and buses in service)	DSAT quarterly and monthly transport statistics (PDF); the source string is recorded row by row in bus_monthly_2020_2026.csv	Cleaned; 45 months (Jan 2023 - Sep 2026); daily trips to Sep 2026, buses in service to Mar 2026; cross-checked against the data.gov.mo mirror (Macao SAR Government, n.d.)
3.5	Optional: hotel room supply, as a control variable	DSEC: https://www.dsec.gov.mo/en-US/	Optional
3.6	MGTO district-hour visitor headcount (24 districts x 24 hours)	Macao Tourism Data Plus: https://dataplus.macaotourism.gov.mo/ (district and hour indicator; indicator page recorded by the M3 lead)	Cleaned; 25,920 rows, 24 districts x 24 hours x 45 months (Jan 2023 - Sep 2026); drives Figures 3 and 5; the zone-level and hourly splits in Figures 2 and 3 are drawn on the seven 2026 months, January to July
3.7	DSAT stop-level scheduled departures (route, stop and hour)	DSAT quarterly and monthly transport statistics (PDF) plus the stop timetable export recorded in supply_stop_departures.csv	Cleaned; 586 stop records, one per stop and direction, carrying scheduled weekday, weekend and holiday departures from the current DSAT timetable rather than a monthly span; joined to the MGTO districts through bus stops; drives Figures 2 and 3

Planned data cleaning steps

- Download the raw tables and keep an untouched copy in the shared repository.
- Write a data dictionary for each dataset (variable name, unit, definition, source, download date).
- Standardize column names and convert month labels to datetime with pandas to_datetime; set a month-end or month-start index consistently.
- Check data types, remove thousand separators and footnote symbols, and convert text numbers to numeric values.
- Check missing values, duplicates, and unusual values; record every decision in a cleaning log.
- Group source markets into Mainland China, Hong Kong, Taiwan, and Others.
- Filter to 2023-01 to 2025-12 (36 months) and merge the datasets on month with pandas merge; check that the merged table has exactly 36 rows, then roll the same table up to 12 quarters, because the seasonality question is asked by quarter.
- Create helper columns: year, month, quarter, and event flags (Lunar New Year, Labour Day, summer, Golden Week).
- Check units and definitions across sources (for example, whether a monthly total is a sum or an average).

The exported panel has one row per month over the whole published span, 45 months from January 2023 to September 2026, plus a twelve-row quarterly roll-up and a district-by-hour layer that runs from January 2023 to September 2026. The zone-level and hourly splits drawn in Figures 2 and 3 are built on the seven 2026 months from January to July, and each caption states its own window. The 36-month window is a filter applied to that panel, not a separate file, so every number in Section 4 traces back to the same rows. Each row keeps year, month, quarter, event flags and the source of every value, so the window and the exclusion rule can be audited from the file itself rather than from this document.

- Export a cleaned master dataset as CSV.

4. Project Tasks and Deliverables

Tasks

No.	Task	Method / tool	Main output
T1	Acquire and clean data (M1 arrivals, M2 hotel, M3 bus)	pandas	Cleaned datasets and data dictionaries
T2	Merge into a master monthly dataset	merge, groupby	Master dataset (36 monthly rows, 12 quarterly rows)
T3	Exploratory data analysis	describe, line plots	EDA figures
T4	Long-run trend chart, 2015-2026	Matplotlib	Figure 1: pandemic gap with the 36-month core window
T5	Synchrony view for RQ1	Matplotlib twinx, axvspan	Twin-axis time series with shaded holidays
T6	Correlation analysis for H1-H3	Seaborn heatmap, scipy.stats.pearsonr, partial correlation	Heatmap and correlation table
T7	Seasonality for RQ2	Quarterly boxplot; seasonal index to remove trend	Boxplots by quarter
T8	Overnight vs same-day distributions	KDE	KDE comparison figure
T9	Basic interactive dashboard	Plotly Dash	Plotly Dash dashboard prototype
T10	Report, presentation, review	Word/PDF, Jupyter Notebook	Analysis report

Tools and techniques

The toolchain is fixed now so that the four weeks are spent on analysis rather than on setup. Every tool below is free, runs on the team's own laptops, and writes an output that a second member can reproduce from the repository.

Tool	Used for in this project	Output
Python 3.11 and pandas 3.1	Reading the CSV and Excel exports, converting text numbers to numeric, merging on month, groupby and pivot for the monthly and quarterly panels	Cleaned datasets, 36 monthly rows and 12 quarterly rows
Matplotlib 3.11	Every static figure, drawn from a colourblind-safe palette, with direct labels, marker and line-style redundancy, and a stated source	Six colour figures at 200 dpi
seaborn and SciPy	Correlation heatmap, Pearson and partial correlation, kernel density estimate	Correlation table and the overnight / same-day distribution figure
Plotly Dash (Plotly Technologies Inc., 2015)	Dashboard prototype: one shared time axis, a district filter and an hour slider	Interactive prototype running from the local files
Jupyter Notebook	One notebook per module, executable from top to bottom without manual edits	Reproducible analysis record handed in with the report
Word and Excel	Report layout, data dictionaries, minutes and the task board	Report, dictionaries, minutes
Git and a shared folder	Version control, branch per module, tagged submission	Repository with the tagged final state

Preliminary analysis of the integrated data

Integration is far enough advanced to show the first result for each research question. The figures below are computed from the cleaned files in the repository rather than from illustrative values. Every coefficient is estimated on the 36 months from January 2023 to December 2025; Figures 2 and 3 are the only panels whose estimates rest on 2026 months, because they read the spatial and hourly splits on the seven months available, January to July 2026, and their captions say so; Figure 1 carries 2026 on the timeline for context only, and no 2026 month enters any statistic here. Every figure uses a colourblind-safe palette in which each series is also distinguished by marker and line style, so it can be read in colour and in black and white.

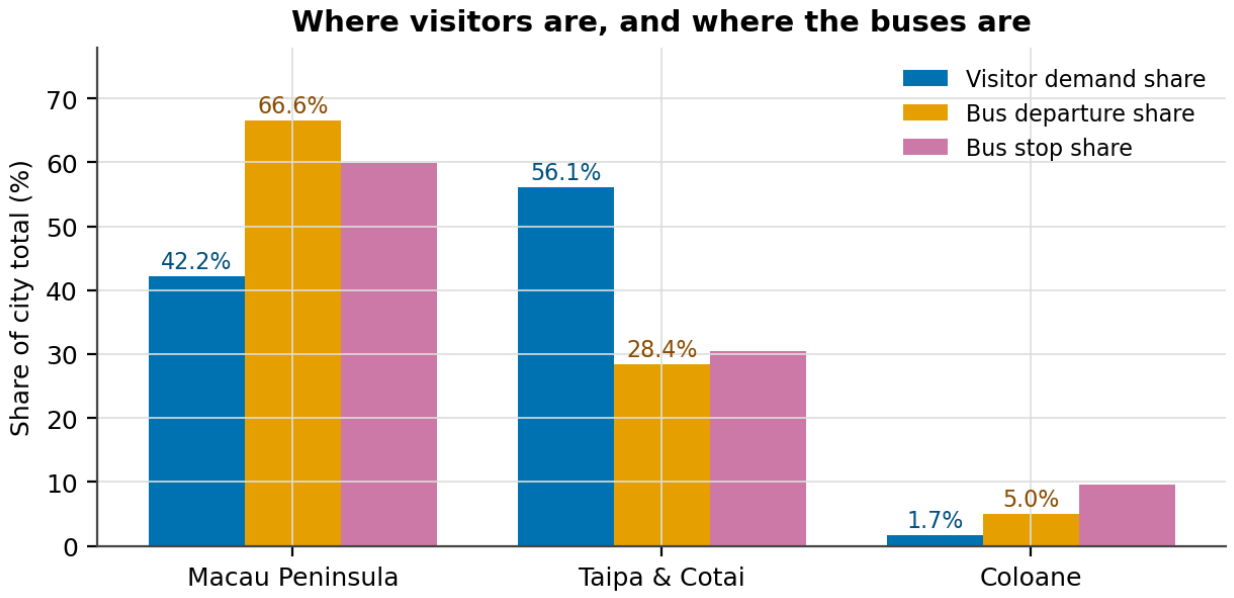


Figure 2. Three shares read on one axis for the first seven months of 2026, January to July: headcount, scheduled departures, and the count of stops. Each bar carries its own percentage, so the blue-orange gap reads directly as the imbalance this project quantifies. Source: MGTO district-hour data (Macao Government Tourism Office, n.d.) and DSAT stop-level departures (Transport Bureau, n.d.); team calculation.

The spatial mismatch is the strongest result so far. Taipa and Cotai receive 56.1 per cent of visitor headcount but only 28.4 per cent of scheduled departures, a supply-to-demand ratio of 0.51, while the Peninsula carries 66.6 per cent of departures for 42.2 per cent of headcount (ratio 1.58). Put differently, the ratio of cumulative visitor presence over January to July 2026 to the daily count of scheduled departures is 204 in Taipa and Cotai against 66 on the Peninsula, so a single departure there absorbs three times the load of one on the Peninsula.

Coloane is over-served on paper (ratio 2.99) but carries 1.7 per cent of headcount, so its surplus capacity is small in absolute terms.

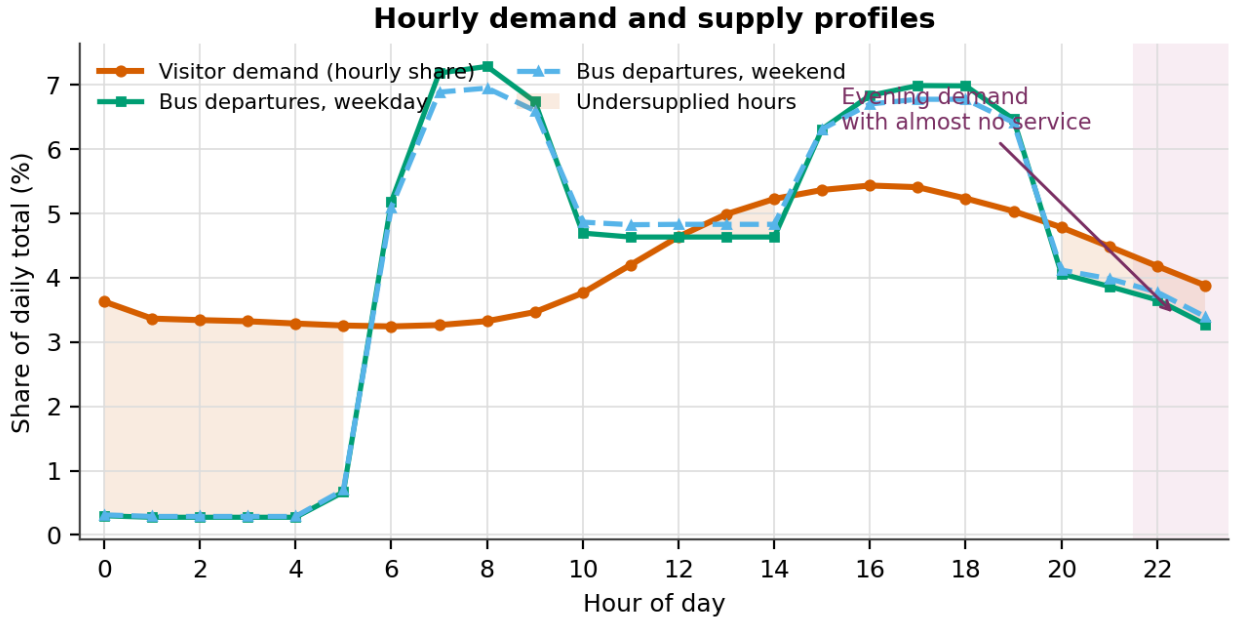


Figure 3. A day-long comparison of crowd and service, built on the same seven months as Figure 2, January to July 2026, so the two panels share one window. The pale fill between the red and the green curves marks the clock times in which scheduled service falls below the measured crowd; the purple strip isolates the late-evening stretch. Each value is a percentage of a full day, and the two service series differ by type of day. Source: as Figure 2.

The temporal mismatch runs the other way. Visitor activity peaks at 16:00 with 5.4 per cent of the daily total, but it also keeps 20.2 per cent of its volume between midnight and 05:59, when only 2.1 per cent of weekday departures are scheduled. Supply concentrates in the middle of the day and thins out exactly when the late-evening visitor needs it, which is the pattern RQ2 is meant to test.

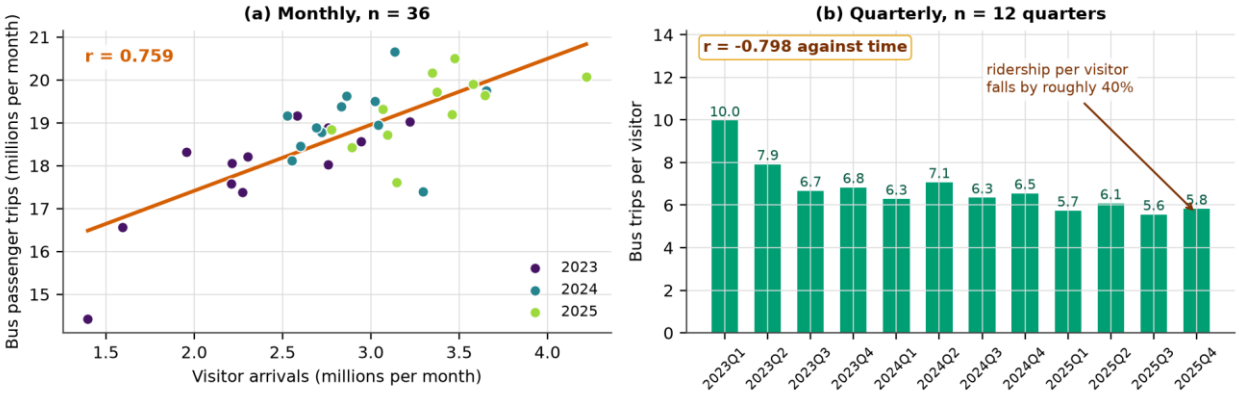


Figure 4. Panel (a) scatters one point per month, one colour per calendar year, with the least-squares line drawn through them and the monthly coefficient printed on the panel; panel (b) gives the intensity ratio for each quarter, prints its own coefficient against time (r = -0.798), and marks its roughly 40 per cent slide with an arrow. Each panel carries its own n, and the quarterly coefficient between arrivals and trips (r = 0.917) is quoted in the text rather than drawn here. Source: Statistics and Census Service (n.d.) for arrivals and occupancy, DSAT monthly bus statistics (Transport Bureau, n.d.) for departures; team calculation.

For RQ1 the window supports H1, with a qualification that matters for the dashboard. Pooled over the twelve quarters, arrivals and passenger trips move almost in step (r = 0.917); pooled over the 36 months the coefficient is 0.759. Inside single years it falls to 0.818 in 2023, 0.254 in 2024 and 0.625 in 2025, and to 0.470 for 2024-2025 together. The reading is straightforward: supply is a capacity decision that grows slowly, while demand

swings with the calendar, so the two series share a level trend more than a month-to-month elasticity. Quoting only the pooled coefficient would overstate how fast the network can respond, which is why the dashboard will show the level view and the deviation-from-trend view side by side.

Splitting arrivals into overnight and same-day visitors sharpens H2 on the volume side only. Over the 36 months, overnight arrivals track hotel guests at r = 0.921 against 0.694 for same-day arrivals, while the occupancy rate moves with both at 0.861 and 0.891, so the hypothesised asymmetry is present in guest volume and not, at this stage, in the occupancy rate; the report will state the result in that form rather than claim the stronger version.

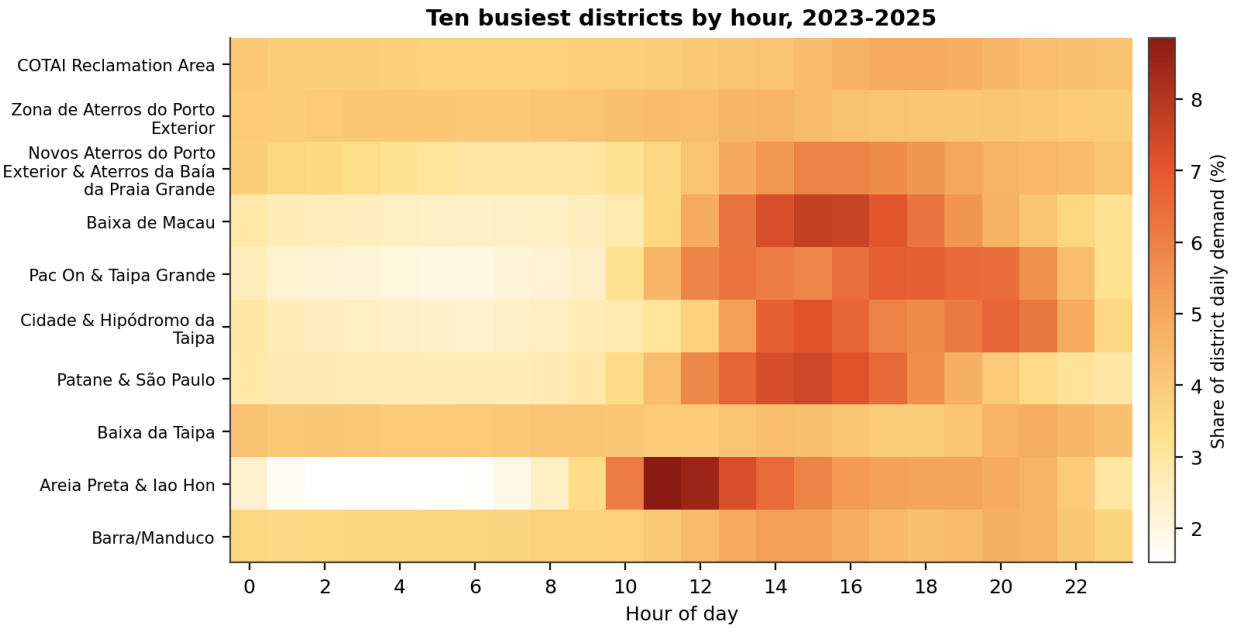


Figure 5. One row for each of the ten busiest districts, each rescaled to that district's own total for a day over the 36-month window, so the rows compare shape rather than volume; darker cells are busier clock times. Rows are the ten districts with the highest total headcount, ordered by that total (the other districts are left out so that the rows stay legible at print size), and the ramp stays ordered when printed without colour. The city-wide shares quoted in the text, 43.7 per cent in Cotai and about 2.1 per cent in an average cell, are read from the un-normalised layer of which these rows are rescaled copies. Source: MGTO district-hour data (Macao Government Tourism Office, n.d.); team calculation.

At the finest resolution the concentration is sharper still. The Cotai reclamation area alone accounts for 43.7 per cent of city-wide headcount over the 36 months, and its 17:00 to 19:00 cells each hold about 2.1 per cent of the city total on their own, roughly four times the busiest cell of any other district. That is the layer that becomes the dashboard's heat surface, and the reason the dashboard needs a district filter rather than a single city-wide view.

Optional / stretch tasks

The following tasks are optional. We will do them only if time allows, and they are not required: a waffle chart for source-market composition; an optional Folium map layer shown inside the Dash dashboard; lag correlation; and OLS regression.

Visualization principles and chart choice

Question type	Chart	Reason
Comparison over time	Twin-axis line chart	Shows co-movement; axes are labelled clearly to avoid misleading scale effects
Relation	Correlation heatmap, scatter plot	Shows direction and strength
Composition	Stacked bar (optional waffle)	Shows source-market share
Distribution	KDE, quarterly boxplot	Compares spread and typical values

We will apply Tufte's ideas (high data-ink ratio, lie factor close to 1) and Cairo's view that charts should be truthful, functional, and insightful. Twin-axis charts can mislead, so each axis will have clear units, and we will also show a standardized view (z-scores) to support the comparison.

Deliverables

- Cleaned master dataset (CSV) with data dictionary.
- Set of figures (PNG), each with title, units, source, and caption.
- Analysis report.
- Interactive dashboard prototype in Plotly Dash, with one shared time axis, a district filter and an hour slider.
- Presentation slides for the Week 4 rehearsal and the final check.
- Five-minute introductory video (Ray lead, Qilan support).
- Meeting minutes for each of the five meetings, circulated within 24 hours by the Scribe.
- Optional: Jupyter Notebook for reproducibility.

5. Success Criteria

The detailed rubric will follow the course requirements and the team work plan. The project is successful if the following criteria are met.

Area	Criterion
Scope and length	This proposal is within 15 A4 pages; the final report follows the course page and format rules.
Data quality	All three indicators are cleaned, documented and merged into 36 monthly rows and 12 quarterly rows on the pandemic-free window; every cleaning step is logged.
Reproducibility	Another team member can re-run the code from raw files to the final figures.
Visualization quality	Each figure has a title, axis labels with units, legend, source note, and readable fonts; no truncated or misleading axes; low clutter; colour-blind-safe palette.
Chart fit	Each chart type matches its question (comparison, relation, composition, distribution).
Statistical care	Correlations are reported with r, its window and its n; the small sample and trend are stated; no causal claims.
Answering questions	RQ1, RQ2, and H1-H3 are each addressed clearly, whatever the outcome.
Honesty	Limitations are stated; the bus data source is documented.
Teamwork	All members contribute to data work and writing; minutes are issued within 24 hours; deadlines are met.
Citations	At least 7 sources in APA 7 style, all verifiable.

Acceptance tests we will run before submission

- One window, stated everywhere: every coefficient is computed on the 36 months from January 2023 to December 2025, its n is printed beside it, and the monthly and quarterly versions agree in sign and magnitude.
- No pandemic month survives in the data: zero rows of the master table fall inside 2020-2022, and the exclusion rule lives in the data dictionary rather than in a chat log.
- The quarterly roll-up reconciles: the twelve quarterly rows sum back to the 36 monthly rows for arrivals, guests and trips, to the last published unit.
- Figures survive both media: each one is checked on screen in colour and in grayscale print, no two series are separated by hue alone, and no label is smaller than 7.5 points.
- Every quoted number is reproducible: the notebook behind each figure runs from top to bottom on a clean environment, and the dashboard opens from the local CSV files in under five seconds without network access.

6. Team Collaboration

Team: Class E - Team 01 (6 members).

Role	Member(s)	Main duties
Team Coordinator	SONG ZHUOXI	Organizes meetings and agendas, keeps the team focused, supports fair participation, resolves blockers
Project Tracker (progress tracker)	KONG KIN MENG, SONG ZHUOXI; (support)	Maintains schedule, milestones, task board, and risk log; alerts the team when behind plan
Team Scribe	FONG HIO LAM; XIAN QILAN (support)	Records minutes, decisions, and action items; stores files in the shared repository
Team Liaison	XIAN QILAN; FONG HIO LAM (support)	Main contact with the instructor; brings feedback back to the team
Project Analyst	KONG KIN MENG, KUAN CHI KIT, FONG CHIN WAI, SONG ZHUOXI, XIAN QILAN	Refine questions, prepare data, analyse, and check conclusions
Project Developer	FONG CHIN WAI, FONG HIO LAM, KONG KIN MENG	Build pipelines, notebooks, visualizations, dashboard, repository
Quality Reviewer	KUAN CHI KIT; FONG CHIN WAI (support)	Check data quality, reproducibility, calculations, visual clarity, citations
Team Reporter	KUAN CHI KIT	Integrate contributions into the report and presentation; coordinate the rehearsal, final check and submission

Data module responsibilities

Module	Lead	Support	Tasks
M1 Visitor arrivals	KONG KIN MENG	SONG ZHUOXI	Acquisition, cleaning, dictionary, baseline time series, holiday/event calendar
M2 Hotel occupancy	KUAN CHI KIT	XIAN QILAN	Acquisition, cleaning, dictionary, occupancy baseline, source-market profile
M3 Bus ridership	FONG CHIN WAI	FONG HIO LAM	Source confirmation, acquisition, cleaning, dashboard-ready tables
Master dataset	FONG CHIN WAI (pipeline)	KONG KIN MENG (validation)	Merge three modules by month
Relationship analysis	KUAN CHI KIT	KONG KIN MENG, FONG CHIN WAI	Correlation and seasonality
Visualization and dashboard	FONG CHIN WAI	FONG HIO LAM (dashboard), KONG KIN MENG	Static figures and the Plotly Dash dashboard prototype
Interpretation	XIAN QILAN	SONG ZHUOXI	Interpretation and storyline

Section ownership: Section G (Data Analysis and Findings) is owned by FONG CHIN WAI; Section I (Visualizations) is owned by XIAN QILAN.

Working rules

- Weekly online meeting on Monday at 20:00; a 15-minute sync on Friday at 17:00.
- Minutes are circulated within 24 hours using the team template; a rotating reviewer checks action items.
- All code, data, and documents are stored in one shared repository with clear file names; raw data is never overwritten.
- Each member holds at most three roles; each report section has one named owner.

- Members report blockers early; the Tracker raises any delay at once.
- Every figure and number is checked by a second member before it enters the report.
- Decisions are made by discussion first, then majority vote if needed, and recorded by the Scribe.

7. Timeline

The project starts on 5 November 2026 and finishes within four weeks, by 3 December 2026.

Week / dates	Focus	Owner(s)	Exit check
Week 1: 5-11 Nov	Acquire and clean M1/M2/M3; data dictionaries; confirm bus data source	Lucas, Ray, Vivi (with support)	Three cleaned datasets and dictionaries in the repository
Week 2: 12-18 Nov	Merge master dataset; EDA; annual context chart	Vivi, Lucas; all analysts	Master dataset (36 months, 12 quarters) and EDA figures reviewed
Week 3: 19-25 Nov	Twin-axis charts, heatmap, Pearson/partial correlation, quarterly boxplots, KDE; Plotly Dash dashboard prototype	Ray, Lucas, Vivi, Sonia	Analysis outputs and dashboard demo
Week 4: 26 Nov-3 Dec	Report draft, narrative, rehearsal; quality review and reproducibility check	Ray, Qilan, Song, Sonia; all	Final package ready by 3 Dec

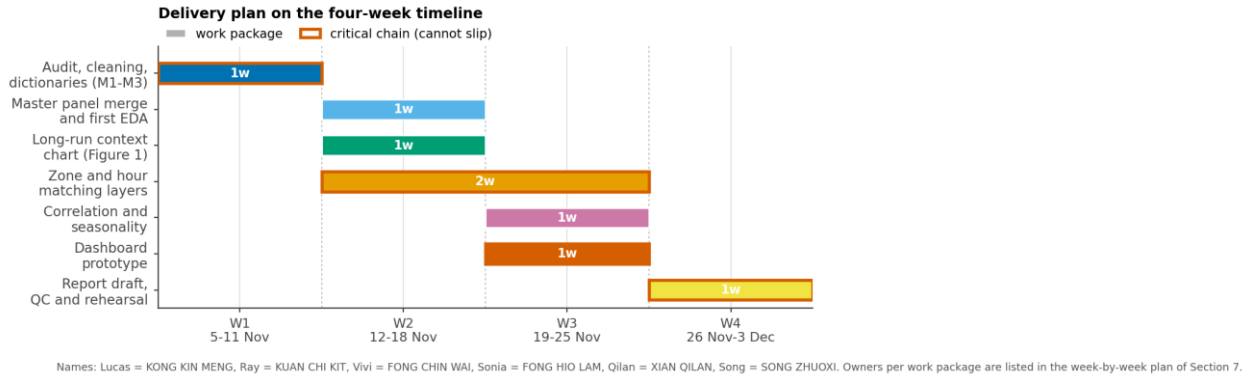


Figure 6. Each bar is a work package placed on the four-week delivery plan of Section 7, its length the planned duration and its label the number of weeks; the red outline marks the chain that cannot slip without moving the delivery date, and the members accountable for each package are named in the week-by-week plan of Section 7. Week 1 starts on 5 November 2026 and the last bar ends on 3 December 2026. Source: team work plan.

The owners in Figure 6 follow the module allocation and the week-by-week plan of Section 7; the overlaps are deliberate, because cleaning must finish before the static figures begin and the dashboard needs the spatial and temporal layers first. The critical path runs from cleaning through the matching layers and the dashboard to the report, so a two-day slip in cleaning consumes the buffer, which is why the risk table treats a late data source, not a late writer, as the main threat.

Meetings: 9 Nov, 16 Nov, 23 Nov, 30 Nov (Mondays, 20:00) and 3 Dec (final check, time to be agreed). Note: the course work plan lists the final report deadline as 5 December 2026; the last two days are kept as a buffer for submission.

Main risks

Risk	Plan
Bus data source not found or incomplete	Confirm the source in Week 1; if monthly data is missing, discuss alternatives with the instructor through the Liaison
Definitions differ across sources	Record definitions in the data dictionary and check units
Delay in one module	Tracker raises it in the Friday sync; support member helps

Misleading twin-axis charts	Add standardized (z-score) view and clear axis labels
Bus data missing or published later than the monthly panel	Fall back on the annual figures plus the district-hour panel, and state the data gap and its limits explicitly in the report

GenAI Use Statement

Tool used: Perplexity AI. It was used to help draft the structure of this proposal and to improve the English wording. The team checked and edited the content, and is responsible for it. The AI did not produce any data or analysis results. All references were selected from real, published sources and should be checked by the team before final submission. For this revision an AI assistant was also used to draft the plotting code behind Figures 1 to 6 and to tighten the English of the added sections; the team re-ran every script on its own data files, checked each figure against the source tables, and corrected the labels, units and numbers itself. All data are taken from the official sources cited below, and no data were generated by a model.

References

Note. The data sets listed below were downloaded from the official portals on 4 November 2026; the archived files, their per-file provenance index and the access dates are kept in the project repository.

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